

AAE 451B – A4

Team 01

9/6/2026



School of Aeronautics
and Astronautics

Morphological Chart

Component	Option 1	Option 2	Option 3	Option 4
Wings	High Wing	Mid Wing	Low Wing	Biplane
Propulsion	Tractor	Pusher	Double Tractor	Double Pusher
Landing Gear	Tricycle	Tail Dragger	Skis	Bicycle
Fuselage	Cylindrical Tube	Box Tube	Lifting Body	No external
Empennage	Conventional	V-Tail	H-Tail	Swept Tail

Configuration Models

- Below are 4 selected concepts for our vehicle
- The options were chosen so that A, B, C, and D are feasible

Concept	Wing	Propulsion	Landing Gear	Fuselage	Empennage
1	High Wing	Tractor	Tricycle	Cylindrical	Standard Tail
2	Low Wing, Dihedral & Taper	Tractor	Tricycle	Cylindrical	Swept Vertical
3	High Wing, Taper & Camber	Double Tractor	Tail Dragger	Cylindrical	H-tail
4	Biplane	Tractor	Tail Dragger	Oval	Swept Vertical

Feasibility Gate

- The letter "Y" signifies being compliant with the presented limit.
- The letter "N" signifies a lack of compliance to the presented limit.

Concept	Within Span Limit?	Within Weight Limit	Within Power Limit?	Holds Battery + Payload?	Passes Gate?
1	Y	Y	Y	Y	Y
2	Y	Y	Y	Y	Y
3	Y	Y	Y	Y	Y
4	Y	Y	Y	Y	Y

Evaluation Matrix: Criteria

1. Takeoff distance: This is a hard requirement set by the RFP and can vary with aircraft configuration.
2. Ease of assembly: There are multiple requirements within the RFP that relate to the speed in which the aircraft needs to be transitioned from a state of disassembled storage to its flight ready configuration, including the empty weight being assembled in under 5 minutes, and the battery and payload being installed within 2 minutes.
3. Battery access: Circles back to RFP requirement of a hard 2 minute cap on battery installation, as well as the RFP requirement that the battery needs to be able to be accessed and removed without removal of aircraft components.
4. Safe landing: RFP dictates a landing within a soft grass field must be conducted without significant damage.
5. Stable/controllable: We need to ensure that our aircraft can reliably complete the 3 lap flight path presented in the RFP while also providing predictable flight behavior so the external pilot may easily avoid potential obstacles in the form of wildlife and/or bystanders.
6. Cost: Because our budget of \$400 is a hard constraint, using expensive materials or manufacturing processes are out of the question. Factoring in cost for manufacturing allows us to obtain a realistic design compromising material benefits for feasibility.

Evaluation Matrix

Objectives	Related Requirements	Weight	Relative Weight	Concept 1	Concept 2	Concept 3	Concept 4
Takeoff distance	R1	3	0.214	2	2	2	3
Ease of assembly	R11, R12, R29	3	0.214	3	2	3	1
Battery access	R13	3	0.214	2	1	2	1
Safe landing	R5	2	0.143	1	2	2	2
Stable/controllable	R4, R17, R18, R19	1	0.071	2	1	2	3
Cost	R23	2	0.143	2	2	1	1
			Score	2.071	1.714	2.071	1.714
			Rank	1	3	1	4

Evaluation Matrix: Weights

- Takeoff distance was weighted heavily because it is a hard requirement
- Ease of assembly and battery access were also weighted heavily because these can be significantly affected by the configuration selection
- Safe landing was given a medium weight because this can be affected by configuration, but is also affected by pilot ability
- Cost was also given a medium weight because it a strict requirement, but we expect most configurations can stay within budget
- Stable/controllable was given the lowest weight because it can be affected by configuration, but between the concepts we traded, stability will largely come down to the detailed design

Configuration Models: Concept 1

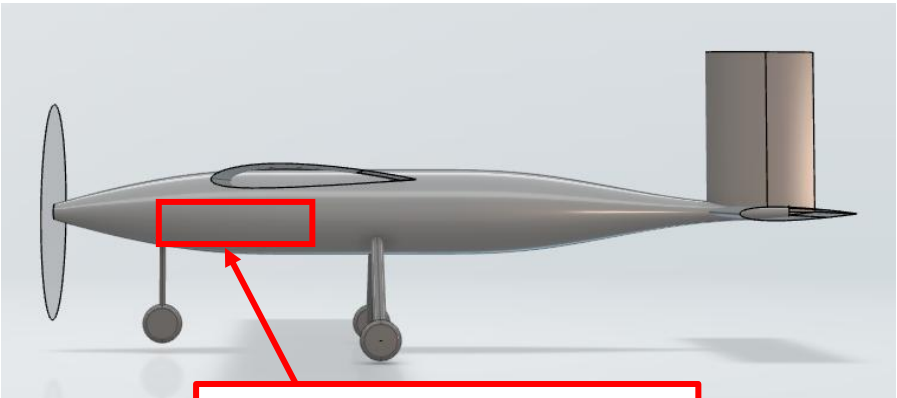
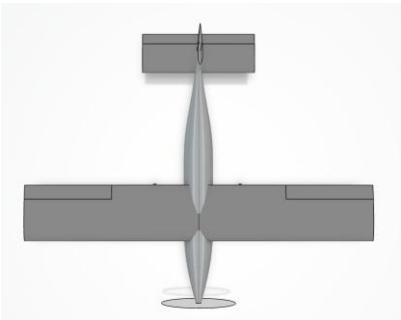
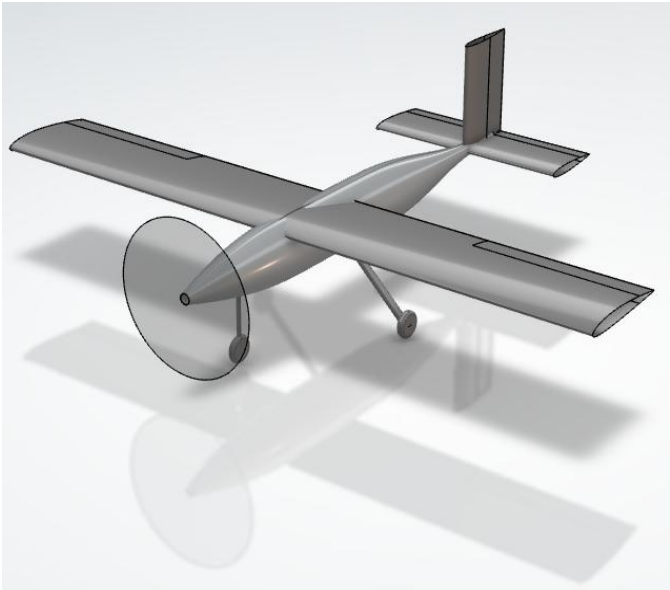
High-Wing, Standard Tail, Tractor

- Consists of a high-wing with a NACA 2412 airfoil
 - Straight wing (no twist or sweep, taper ratio of 1)
- Ailerons that occupy 50% of span and 25% of chord
- Tricycle landing gear - no extra clearance needed for tractor propeller
- Standard tail
- Payload, battery, and primary avionics reside in fuselage cavity
- First row of parameter sheet

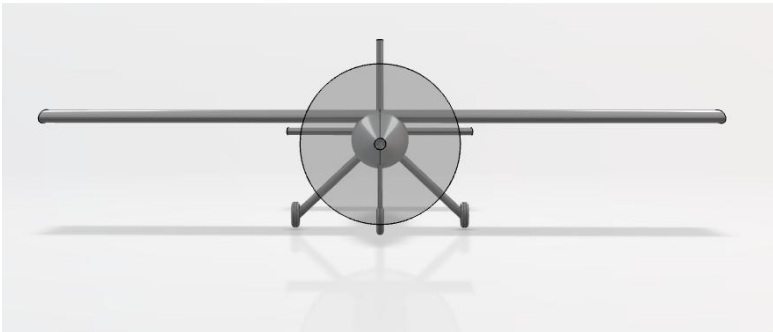
Score Justification

- Takeoff distance: Given a 2 due to the lift C_L , no flaps, and wing area being about average relative to other choices
- Ease of assembly: High wing, single motor, and single rudder make this simple to assembly
- Battery access: Battery could be accessed through either the side or bottom, but not top
- Safe landing: Risk of a tail strike during landing if not careful
- Stable/controllable: High wings are more stable
- Cost: Relatively simple design without duplicated components

Objectives	Concept 1
Takeoff distance	2
Ease of assembly	3
Battery access	2
Safe landing	1
Stable/controllable	2
Cost	2



Payload, Battery, Avionics



Configuration Models: Concept 2

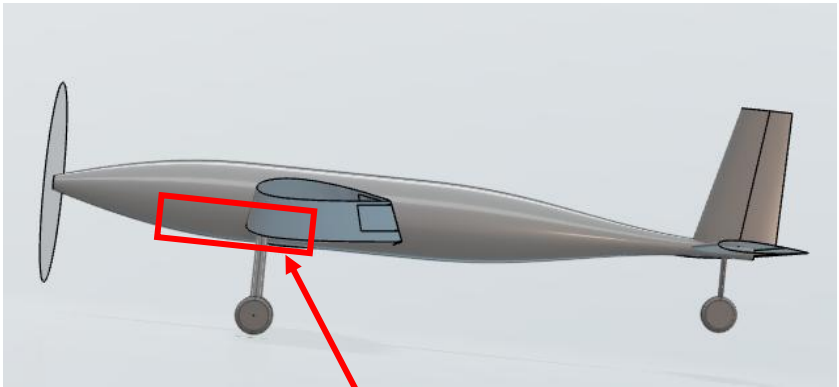
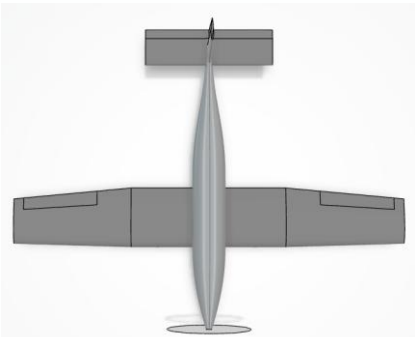
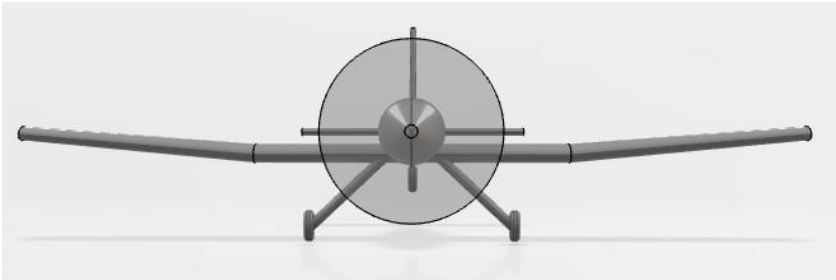
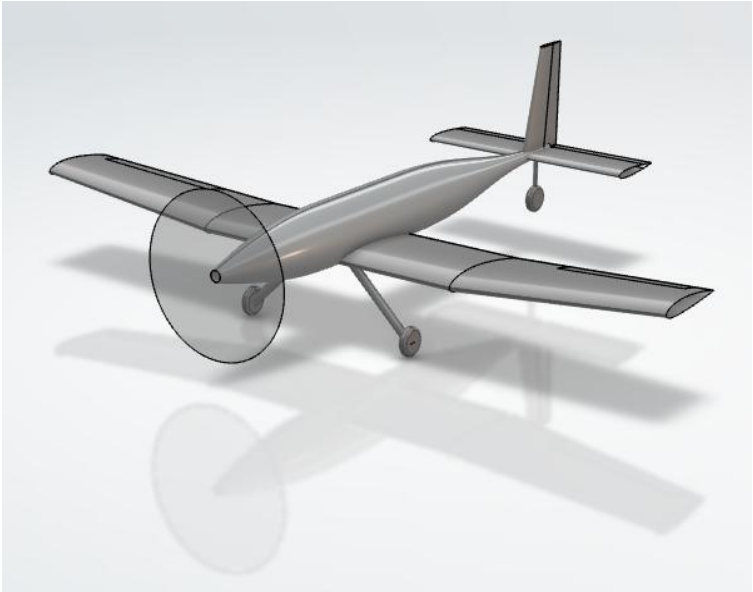
Bent Tapered Low-Wing, Standard Tail, Tail Dragger

- Consists of a low-wing with a NACA 2415 airfoil
 - Straight wing for 40% span, other 60% at 5° dihedral, taper ratio 0.75
- Ailerons only on dihedral part of wing
- Tail Dragger landing gear – gives extra prop clearance for tractor
- Swept Tail at 5° backward with tapered vertical stabilizer
- Payload, battery, and primary avionics reside in fuselage cavity
- Second row of parameter sheet

Score Justification

- Takeoff distance: Given a 2 due to the lift C_L , no flaps, and wing area being about average relative to other choices.
- Ease of assembly: Simple aircraft with minimal components, but due to low wing might be hard to assembly the landing gear, wing, and body with the payload and battery in this area.
- Battery access: Wing/body structure will interfere with desired battery location, possibly making it harder to fit a battery in this area with ease.
- Safe landing: Tail dragger is easier to land because less risk of tail strike, more prop clearance, and twin props decrease prop size and lower strike risk, but easier to flip than tricycle
- Stable/controllable: Propeller adds net torque to vehicle, and swirled flow hits the tail that can cause instabilities.
- Cost: Low overall cost due to minimal material and single engine.

Objectives	Concept 2
Takeoff distance	2
Ease of assembly	2
Battery access	1
Safe landing	2
Stable/controllable	1
Cost	2



Payload, Battery, Avionics

Configuration Models: Concept 3

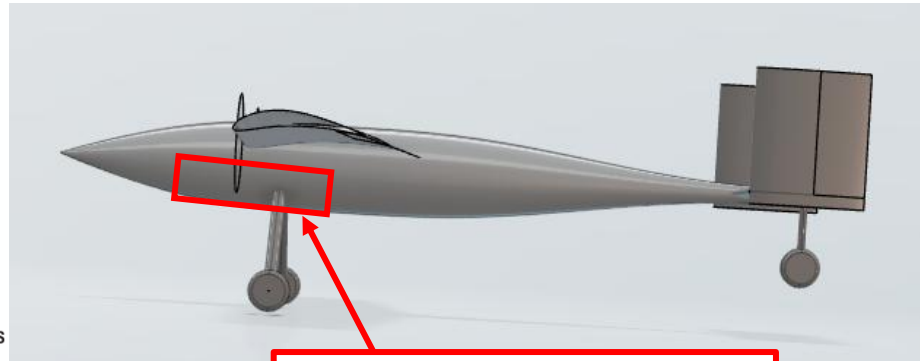
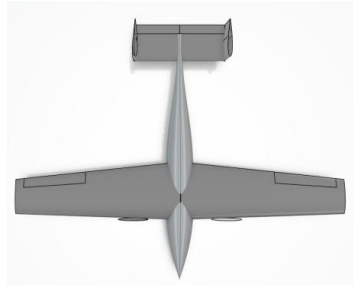
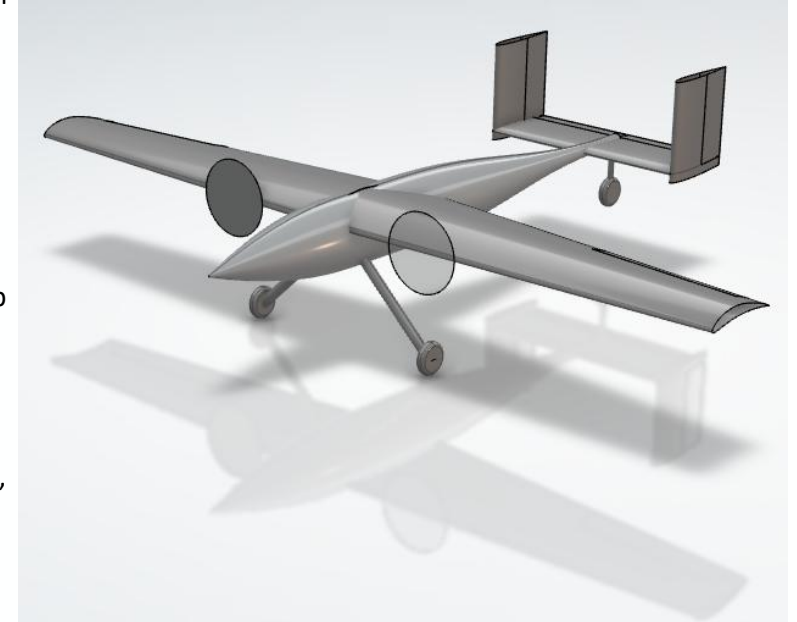
Cambered High-Wing, H-Tail, Double Tractor, Tail Dragger

Objectives	Concept 3
Takeoff distance	2
Ease of assembly	3
Battery access	2
Safe landing	2
Stable/controllable	2
Cost	1

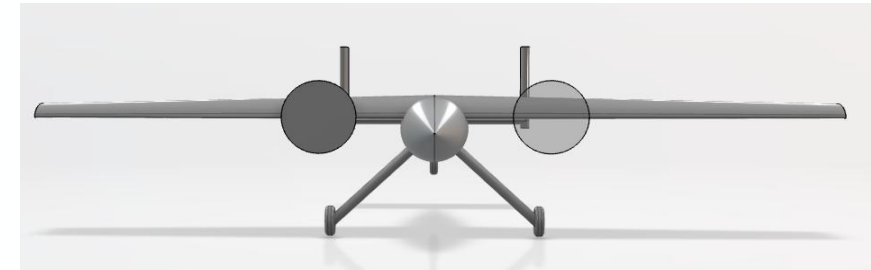
- Consists of a high-wing with E423 airfoil to balance H-tail moment
 - Taper ratio 0.6, no twist or sweep
- Ailerons on only ~38% of wing; double tractor propellers on front
- Tail Dragger landing gear – gives extra prop clearance
- H-tail with smaller span and smaller vertical height than standard
- Payload, battery, and primary avionics reside in fuselage cavity
- Third row of parameter sheet

Score Justification

- Takeoff distance: Given a 2 due to the lift C_L , no flaps, and wing area being about average relative to other choices. Initial AoA from tail dragger but smaller props less efficient
- Ease of assembly: Two motors and two rudders make assembly more complicated
- Battery access: Battery could be accessed through either the side or bottom, but not top
- Safe landing: Tail dragger is easier to land because less risk of tail strike, more prop clearance, and twin props decrease prop size and lower strike risk, but easier to flip than tricycle
- Stable/controllable: Dual engines cancel out torque, and increases stability in low speeds and while turning
- Cost: Dual engines more expensive and require more avionics, also additional tail material brings cost up



Payload, Battery, Avionics



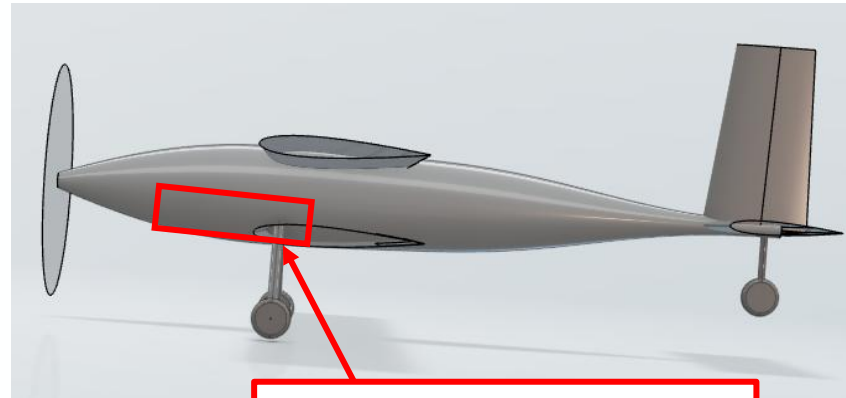
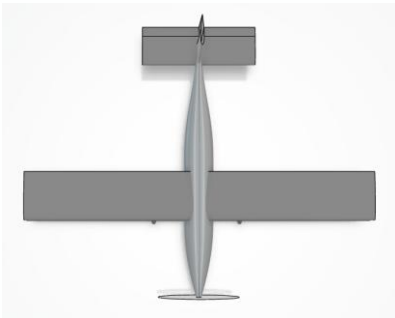
Configuration Models: Concept 4

Biplane, H-Tail, Standard Tail, Tail Dragger

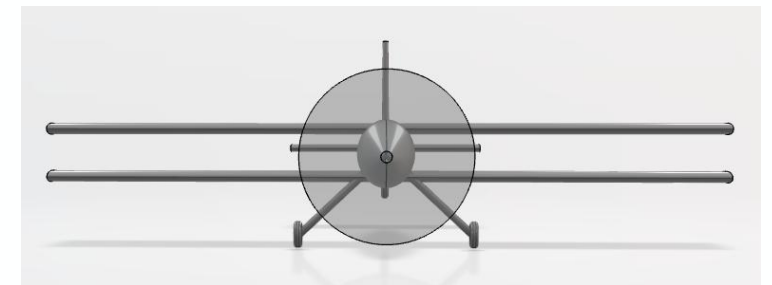
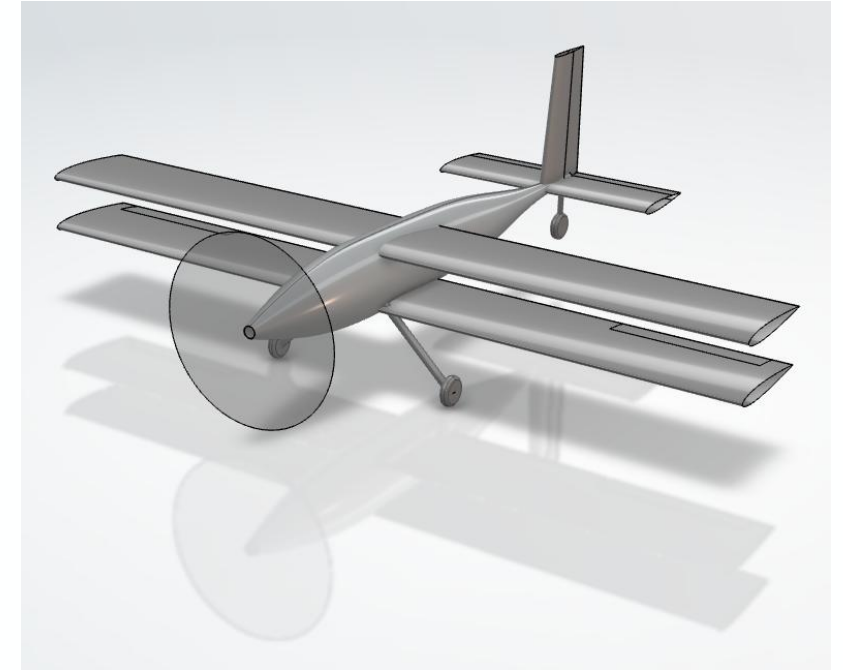
Objectives	Concept 4
Takeoff distance	3
Ease of assembly	1
Battery access	1
Safe landing	2
Stable/controllable	3
Cost	1

Score Justification

- Consists of two wings with NACA 0012 airfoils
 - Straight wings (no twist or sweep) attached to oval fuselage
 - Ailerons on only ~38% of lower wing, none on upper wing
 - Tail Dragger landing gear – gives extra prop clearance
 - Standard tail, slightly swept vertical stabilizer
 - Payload, battery, and primary avionics reside in fuselage cavity
 - Fourth row of parameter sheet
- Takeoff distance: Given a 3 due to the high wing area and better clearance of prop from the ground, as well as a starting AoA from tail dragger gear.
 - Ease of assembly: Increased amount of build and assembly required for 2 wings, location of components within the vehicle become harder to fit.
 - Battery access: Wing/body structure will interfere with desired battery location, possibly making it harder to fit a battery in this area with ease.
 - Safe landing: Tail dragger is easier to land because less risk of tail strike, more prop clearance, and twin props decrease prop size and lower strike risk, but easier to flip than tricycle
 - Stable/controllable: Will likely perform well in low speed, low wing loading makes it more maneuverable
 - Cost: Due to extra wing material, will require more material, bringing cost up.



Payload, Battery, Avionics



Configuration Model Parameters

Main Wing	Config	Break % Span	Dihedral Inner (deg)	Dihedral Outer (deg)	Sweep Inner (deg)	Sweep Outer (deg)	Taper Inner	Taper Outer	Twist @ Break (deg)	Twist @ Tip (deg)	Root Chord (mm)	Full Span (mm)	Wing Incidence (deg)	Vertical Wing Location (mm)	Airfoil @ Root	Airfoil @ Break	Airfoil @ Tip
	1	0.10%	0	0	0	0	1	1	0	0	203	1300	0	40	3	3	3
	2	40%	0	5	0	0	1	0.75	0	0	203	1300	0	-40	4	4	4
	3	0.10%	0	0	0	0	1	0.6	0	0	225	1300	0	20	5	5	5
	4	0.10%	0	0	0	0	1	1	0	0	180	1300	0	-45	2	2	2

Second Wing (Config 4 ONLY)	Root Chord (mm)	Airfoil @ Root	Taper Inner	Airfoil @ Break	Twist @ Break	Taper Outer	Airfoil @ Tip	Twist @ Tip	Dihedral Inner (deg)	Dihedral Outer (deg)	Sweep Inner (deg)	Sweep Outer (deg)	Total Span (mm)	Break % Span	Wing Incidence (deg)	Wing Gap (mm)	Aileron?
	180	2	1	2	0	1	2	0	0	0	0	0	1300	0.10%	0	92	FALSE

Main Aileron	Config	Chord %	End %	Start %	Left Deflection (deg)	Right Deflection (deg)	Active?
	1	25%	100%	50%	0	0	TRUE
	2	25%	95%	57.50%	0	0	TRUE
	3	25%	95%	57.50%	0	0	TRUE
	4	25%	95%	57.50%	0	0	TRUE

Ref. & Positions	Config	Target X.CG (mm)	Wing C/4 Location (mm)	Vert. Tail Moment Arm (mm)	Horiz. Tail Moment Arm (mm)	Wing 2 C/4 Location (mm)
	1	250	280	600	600	N/A
	2	250	330	600	600	N/A
	3	250	290	600	600	N/A
	4	250	330	570	570	330

Horizontal Tail	Config	Root Chord (mm)	Airfoil @ Root	Incidence Angle (deg)	Full Span (mm)	Break % Span	Dihedral Inner (deg)	Sweep Inner (deg)	Twist @ Break	Airfoil @ Break	Dihedral Outer (deg)	Sweep Outer (deg)	Twist @ Tip (deg)	Airfoil @ Tip	Taper Inner	Taper Outer
	1	141	2	0	423	0.10%	0	0	0	2	0	0	0	2	1	1
	2	120	2	0	423	0.10%	0	0	0	2	0	0	0	2	1	1
	3	141	2	0	360	0.10%	0	0	0	2	0	0	0	2	1	1
	4	141	2	0	423	0.10%	0	0	0	2	0	0	0	2	1	1

Configuration Model Parameters

Vertical Tail	Config	Height (mm)	Taper Ratio	Root Chord (mm)	Sweep (deg)	Active?	Tail Config	H-Tail Ratio	H-Tail Lower Taper	Vertical Location (mm)	Airfoil
	1	210	1	141	0	TRUE	1	N/A	1	0	2
	2	200	0.6	120	12	TRUE	1	N/A	1	0	2
	3	150	1	141	0	TRUE	2	0.1	1	0	2
	4	240	0.75	141	5	TRUE	1	N/A	1	0	2

Fuselage	Config	Tail Boom Radius (mm)	Tail Boom Length (mm)	Nose Radius (mm)	Sec. 1 SemiMajor (mm)	Sec. 1 SemiMinor (mm)	Sec. 1 X-Loc (mm)	Sec. 2 X-Loc (mm)	Sec. 2 SemiMajor (mm)	Sec. 2 SemiMinor (mm)	Boom Center (mm)
	1	10	1	10	55	55	300	500	55	55	0
	2	10	1	10	55	55	300	500	55	55	0
	3	10	1	1	55	55	300	380	55	55	0
	4	10	1	10	65	55	300	500	60	55	0

Landing Gear	Config	Main Gear X-Loc (mm)	Main Gear Length (mm)	Main Gear Wheel Thickness (mm)	Main Gear Support Radius (mm)	Aux. Gear X-Loc (mm)	Aux. Gear Length (mm)	Aux. Gear Wheel Thickness (mm)	Aux. Gear Support Radius (mm)	Aux. Gear Wheel Dia (mm)	Main Gear Dia (mm)	Main Gear Track (mm)
	1	415	155	15	8	140	145	12.5	4	50	50	330
	2	300	155	15	8	950	90	15	4	50	50	330
	3	280	155	15	8	950	90	15	5	50	50	330
	4	300	155	15	7	950	90	15	5	50	50	330

Elevator	Config	% Start	% End	Chord %	Left Deflection (deg)	Right Deflection (deg)	Active?
	1	0%	100%	25%	0	0	TRUE
	2	0%	100%	25%	0	0	TRUE
	3	0%	100%	25%	0	0	TRUE
	4	0%	100%	25%	0	0	TRUE

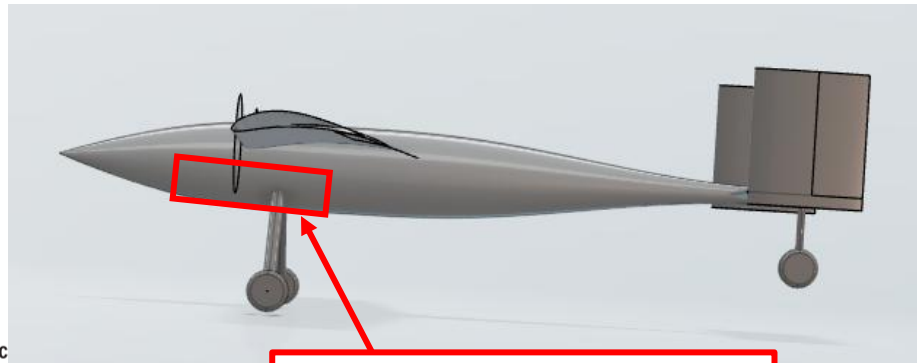
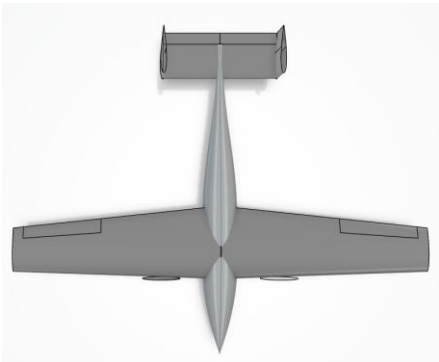
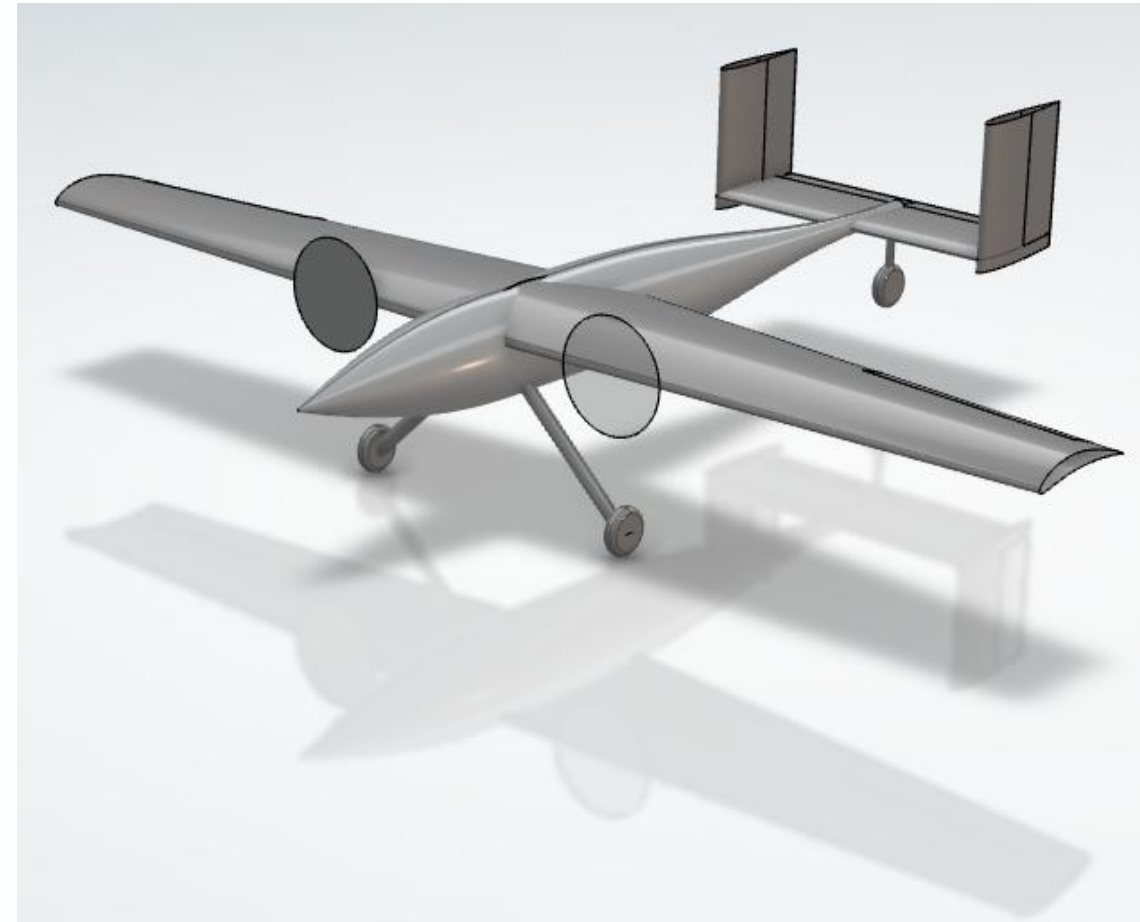
Rudder	Config	% Start	% End	Chord %	Angle (deg)	Active?
	1	0%	100%	45%	0	TRUE
	2	0%	100%	45%	0	TRUE
	3	0%	100%	45%	0	TRUE
	4	0%	100%	45%	0	TRUE

Propeller 1	Config	X-Coord (mm)	Y-Coord (mm)	Z-Coord (mm)	Diameter (mm)	Thickness (mm)
	1	0	0	0	280	20
	2	0	0	0	280	20
	3	-235	185	20	120	20
	4	0	0	0	304.8	20

Propeller 2 (Config 3 ONLY)	Config	X-Coord (mm)	Y-Coord (mm)	Z-Coord (mm)	Diameter (mm)	Thickness (mm)
	3	-235	-185	20	120	20

Final Concept

- Concept 3 has been chosen due to the result of the weighted matrix, even though it was tied for the 1st concept, and may be more expensive than concept 1
- Chosen due to unique design and interesting challenge of using an H tail over conventional tail
- Bi-prop was chosen for easy of controllability due to torque imbalance from the propeller
- Due to high wing design, wing can be a singular piece while still having access to battery and payload
- Tail dragger chosen for lower overall drag and better propeller clearance
- Tapered wing for better spanwise efficiency



Payload, Battery, Avionics

